

Data- Driven Analysis Traffic Crash Causes

Joy Njeru

Objectives & Stakeholders

OBJECTIVES

01

Predict whether a crash results in injury or no injury

02

Identify the most influential contributing factors

03

Translate findings into actionable safety recommendations

STAKEHOLDERS

City Transportation Authority

Policy & resource allocation

Road Safety Agencies

Awareness campaigns

Urban Planners

Infrastructure & road design

METHODOLOGY

Data Sources & Modelling Pipeline

Crashes.csv

Accident-Level

- Date/time & location
- Contributory cause
- Weather & lighting
- Road conditions
- Crash type

Vehicles.csv

Vehicle-Level

- Vehicle type
- Vehicle Age
- Driving maneuver
- Unit type

People.csv

Person-Level

- Person type & sex
- Age
- Safety equipment
- Airbag deployment

MODELLING PIPELINE

Data Cleaning

Aggregation & Merge

Label Encoding

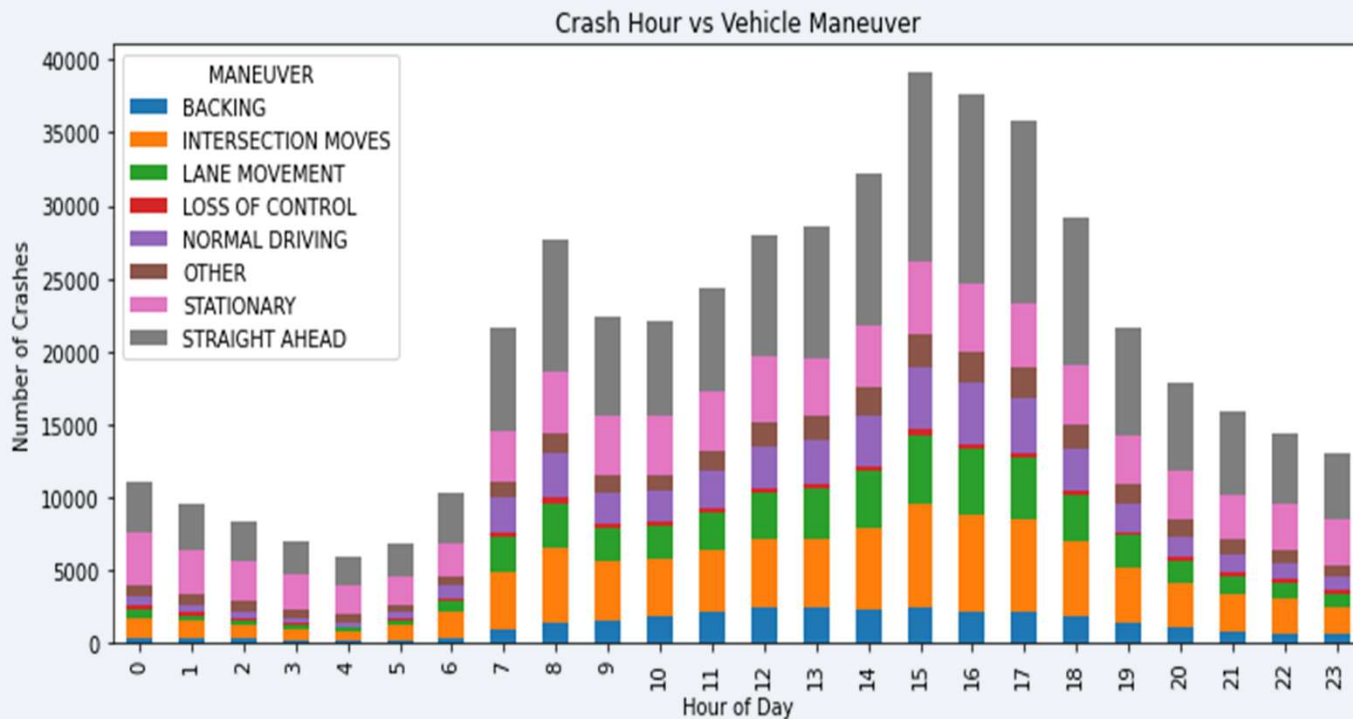
Train/Test Split

SMOTE Balancing

Random Forest

FINDINGS

Crash Frequency by Hour of Day



Early Hours

12 AM – 4 AM

Low volume, high severity per crash due to DUI fatigue, and low visibility

Morning Rush

7 AM – 9 AM

High congestion crash volume; following too closely dominate

Evening Peak

4 PM – 6 PM

Highest crash frequency of the day — commuter volume, fatigue, and diminishing light

RECOMMENDATIONS

Actionable Safety Interventions

01 Public Awareness

- Right-of-way education — #1 crash cause
- Following-distance campaigns for urban commuters
- Targeted messaging by age demographics

02 Emergency Response

- Pre-position units at crash hotspots
- Dynamic resource allocation by time and weather
- Reduce response time to high-severity corridors

CONCLUSION

Traffic crash severity can be meaningfully predicted using integrated crash, vehicle, and people-level data. Human behavior dominates crash causes, while environmental conditions amplify severity.

#1

Cause: Failing to Yield

4–6 PM

Peak Crash Window

0.52

Optimized Model Threshold

By operationalizing these findings, city authorities can transition from reactive incident response to proactive, data-informed crash prevention.